

HP PageWide 300, 400, and 500 series - PQ (Print Quality) Library: PQ Issues Caused by Printer Hardware

Introduction

This document contains examples of print quality defects and information to help you troubleshoot these defects.

To determine which defects are present, print a **Printer Status Report** and a **Print Quality Report**, and verify that the customer is using genuine HP ink. Then compare the reports to the examples in this document. Once you have determined which defect(s) are present, click the link in the table for that defect to skip to the section in this document that contains information for resolving the print quality issue.

IMPORTANT: During the troubleshooting process, each time a customer prints a PQ Report, ask them to make a note on the report of any calibrations or cleaning procedures that they performed *prior* to printing the report.

The Printer Status Report shows the number of calibrations that have been performed, as well as page counts for the last successful and/or failed calibrations.

Use the links below to skip to these sections in this document:

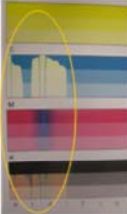
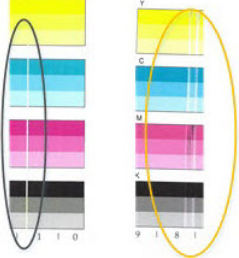
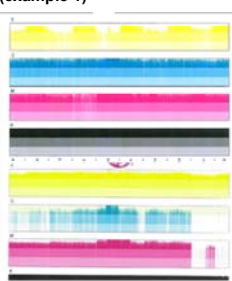
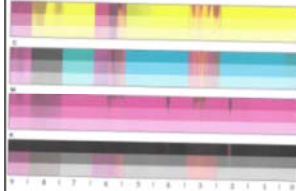
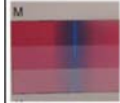
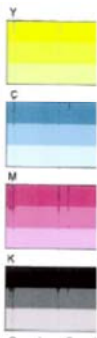
[Obtain a Printer Status Report, Print Quality Report, and run a Tap Test](#)

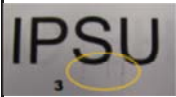
[Identify genuine HP ink and non-HP ink](#)

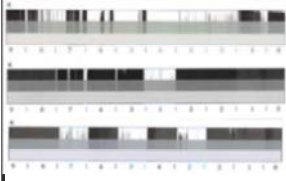
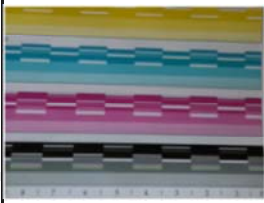
Examples of print quality defects

The three tables below contain examples of different print quality defects.

Once you have determined which defect(s) are present, click the link in the table for that defect to skip to the section in this document that contains information for resolving the print quality issue.

Large sections of wrong or missing color	Small sections of wrong or missing color	Streaks across entire page - Multiple colors improving down the page	Color mix
IMPORTANT Call center agents: Be sure to log the specific print quality defect in your case notes. See the last row in this table for information on how to log the specific defect.			
PQ 1A: Large sections color mixing 	PQ 2A: Streaks across all colors 	PQ 3A: Color improving down page (example 1) 	PQ 4A: Color mixing (example 1) 
PQ 1B: Large sections missing color 	PQ 2B: Streaks across a single color (list the color) 	PQ 3A: Color improving down page (example 2) 	PQ 4A: Color mixing (example 2) 
Call center agents Log as one of the following: - PQ 1A: Large sections color mixing - PQ 1B: Large sections missing color	Call center agents Log as one of the following: - PQ 2A: Streaks across all colors - PQ 2B: Streaks across a single color (list the color)	Call center agents Log as: PQ 3A: Color improving down the page	Call center agents Log as: PQ 4A: Color mixing
Streaks across entire page - Single or multiple colors consistent down the page	Streaks in yellow	Ink smear	Image stretched vertically
IMPORTANT Call center agents: Be sure to log the specific print quality defect in your case notes. See the last row in this table for information on how to log the specific defect.			
PQ 5A: Streaks consistent down the page (example 1) 	PQ 6A: Streaks in yellow 	PQ 7A: Ink smear (example 1) 	PQ 8A: Stretched image 
PQ 5A: Streaks consistent down the page (example 2) 		PQ 7A: Ink smear (example 2) 	

 PQ 5A: Streaks consistent down the page (example 3) 			
Call center agents Log as: PQ 5A: Streaks consistent down the page	Call center agents Log as: PQ 6A: Streaks in yellow	Call center agents Log as: PQ 7A: Ink smear	Call center agents Log as: PQ 8A: Stretched image

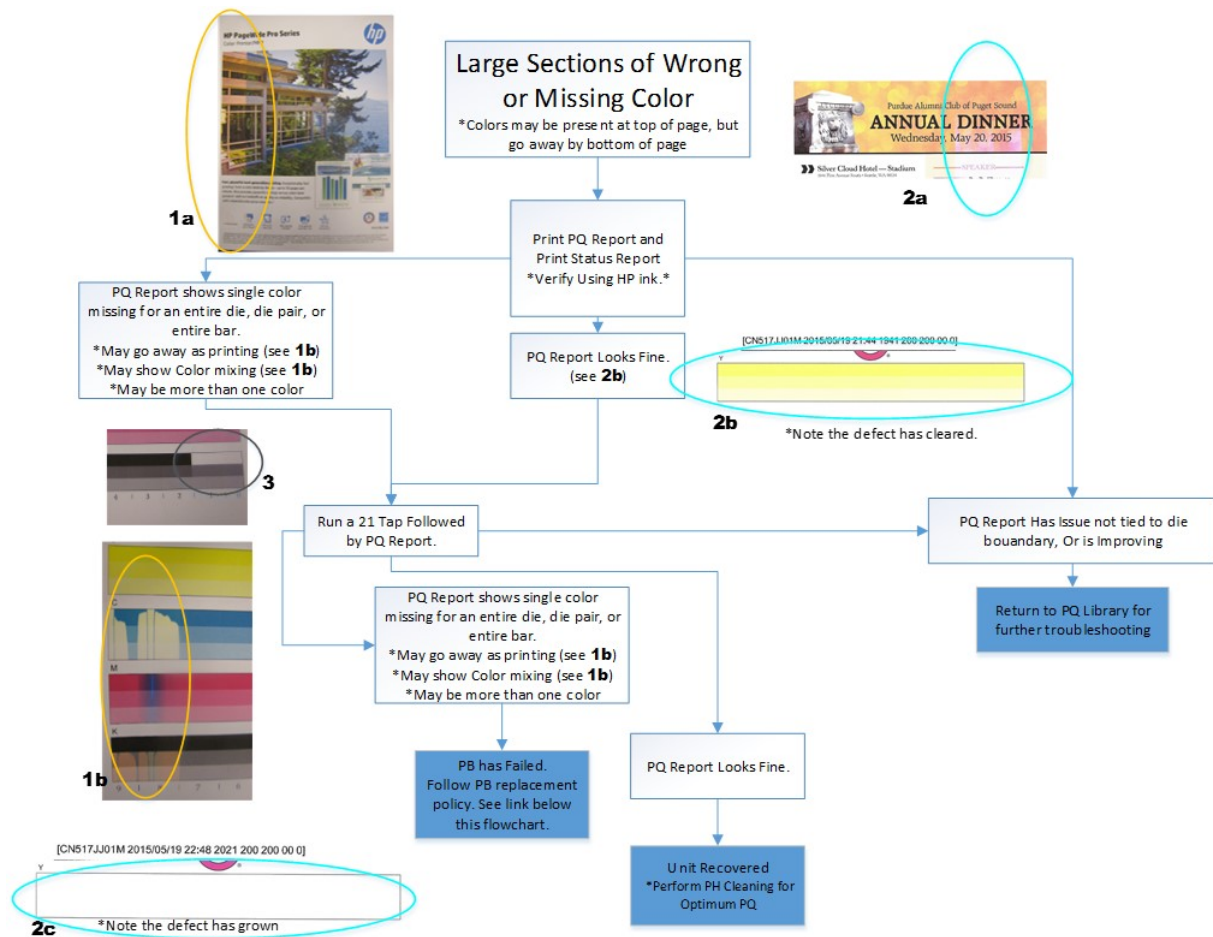
Vertical white bands on one color	Light and dark regions	Columns of misaligned print	White lines in black (on new printer)
IMPORTANT Call center agents: Be sure to log the specific print quality defect in your case notes. See the last row in this table for information on how to log the specific defect.			
PQ 9A: Massive nozzles out with streaks 	PQ 10A: Light and dark regions 	PQ 11A: Misaligned print 	PQ 12A: White lines in black (new printer) 
Call center agents Log as: PQ 9A: Massive nozzles out with streaks	Call center agents Log as: PQ 10A: Light and dark regions	Call center agents Log as: PQ 11A: Misaligned print	Call center agents Log as: PQ 12A: White lines in black (new printer)

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Troubleshooting print quality defects

The following sections contain flow charts to assist in troubleshooting specific print quality defects. After determining the defect that is present on the Printer Status Report and/or the Print Quality Report (using the examples shown in the tables above), click the link provided in the table to skip to the section that contains troubleshooting steps to resolve the specific print quality issue.

Large sections of wrong or missing color



Possible causes:

- **Non-recoverable:** Caused from air in the printbar.
- **Recoverable:** Caused from a singular event that should not reoccur.

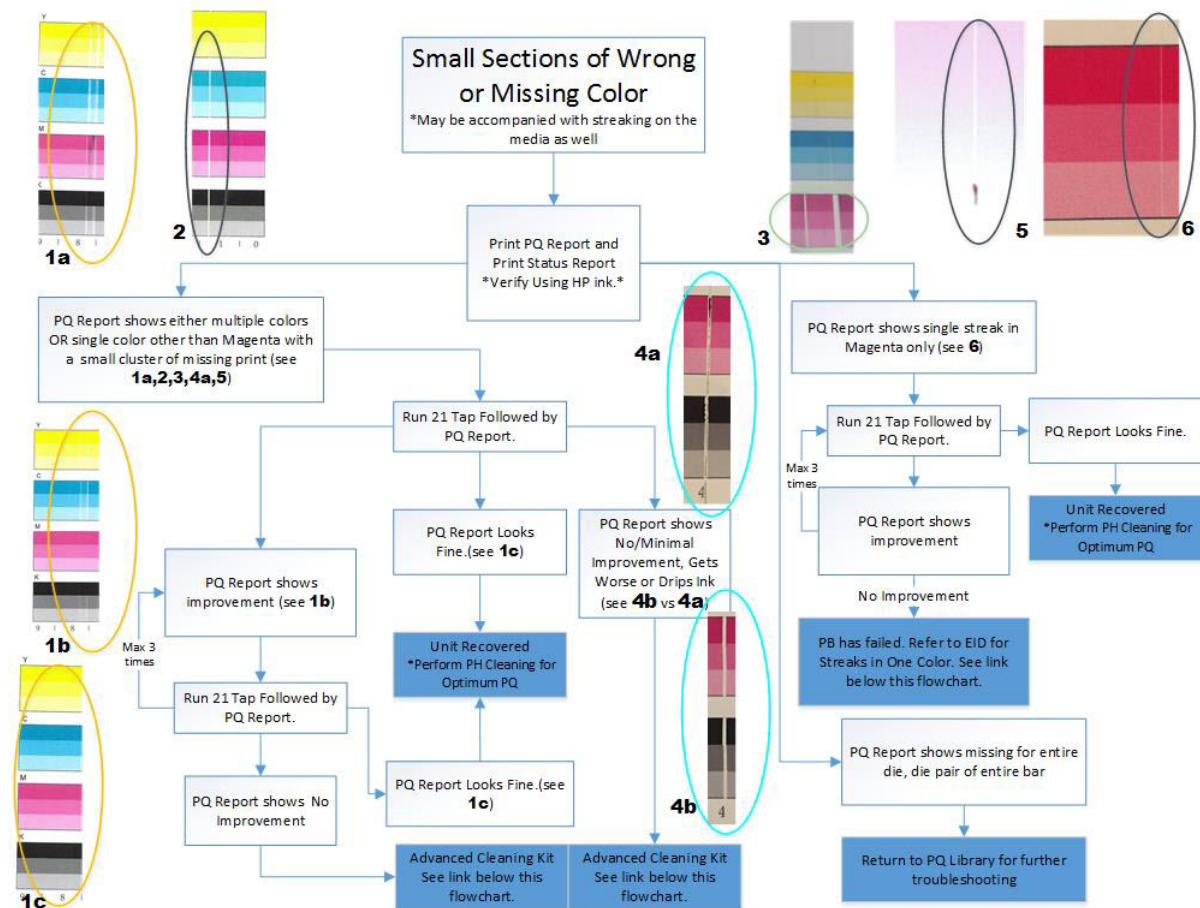
Click any of the following links to skip to the section in this document that contains steps to perform the specific task:

- [Print a Print Quality Report](#)
- [Print a Printer Status Report](#)
- [Run a 21 Tap Test](#)

If the print bar has failed, see the [Printbar Replacement Policy](#) (c04926672) for next steps.

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Small sections of wrong or missing color



Light and dark lines less severe than those shown in this issue could be a result of alignment issues. Refer to [Solve PQ \(Print Quality\) Problems Due to Technology Limitations](#) (c05063184) for steps to resolve alignment issues.

Possible causes:

- **Non-recoverable:** Might have been caused by external contaminants on UPG shims or internal contaminants in the nozzles.
- **Recoverable:** Caused by external contaminants on the die, and should not reappear.

Click any of the following links to skip to the section in this document that contains steps to perform the specific task:

- [Print a Print Quality Report](#)
- [Print a Printer Status Report](#)
- [Run a 21 Tap Test](#)

If the print bar has failed, see [Printouts showing streaks across a single color \(Emerging Issue\)](#) (c05103438).

If the PQ Report shows no/minimal improvement, please order the **HP Advanced Cleaning Kit** (also known as the HP Shim Whisker Kit). The part number for this kit is CN459-67006. The kit contains instructions on how to use it.

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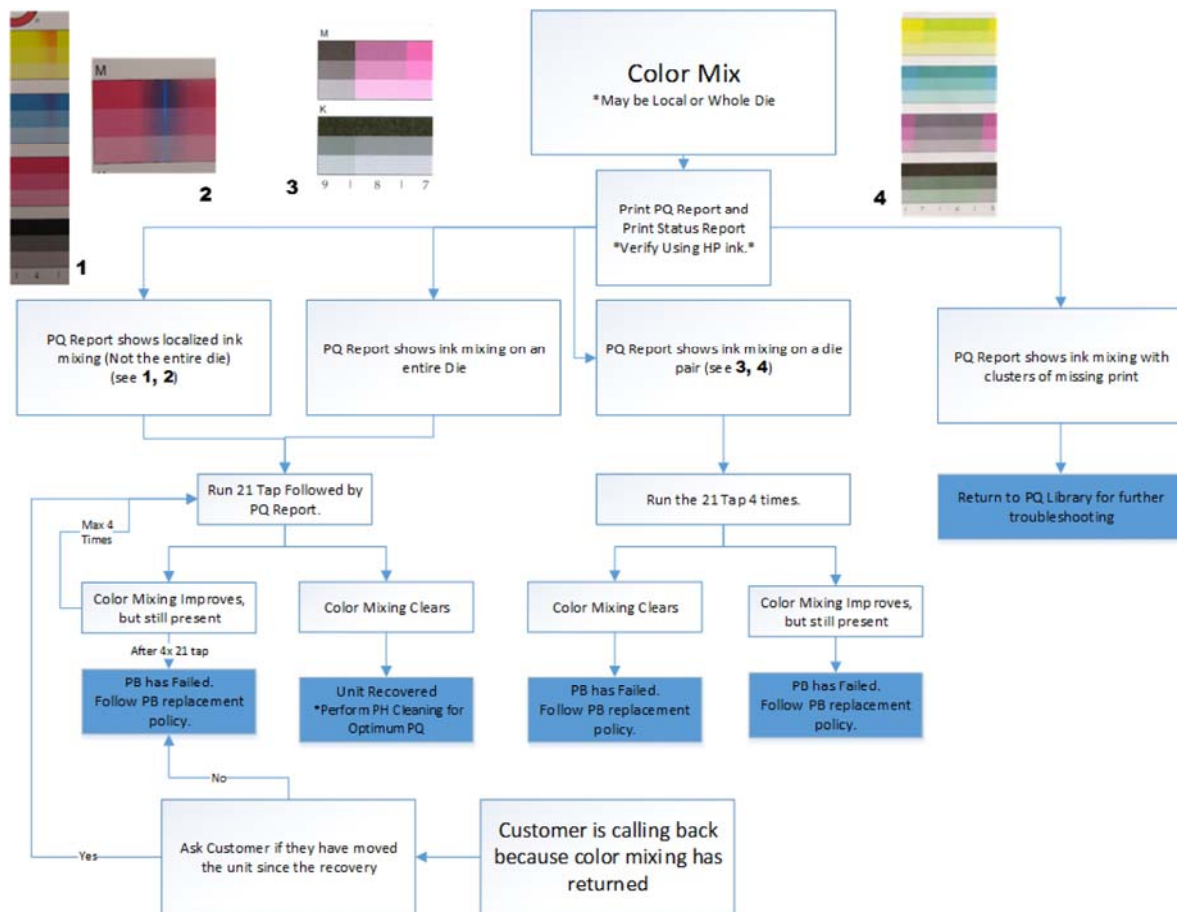
Streaks across entire page - Multiple colors improving down the page

Please refer to the following document for steps on resolving this issue:

[Streaks across entire page - Multiple colors improving down the page \(Emerging Issue\)](#) (c05337092)

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Color Mix

**Possible causes:**

- **Non-recoverable:** Caused by a printbar defect.
- **Recoverable:** Caused by external contamination and should not reoccur.

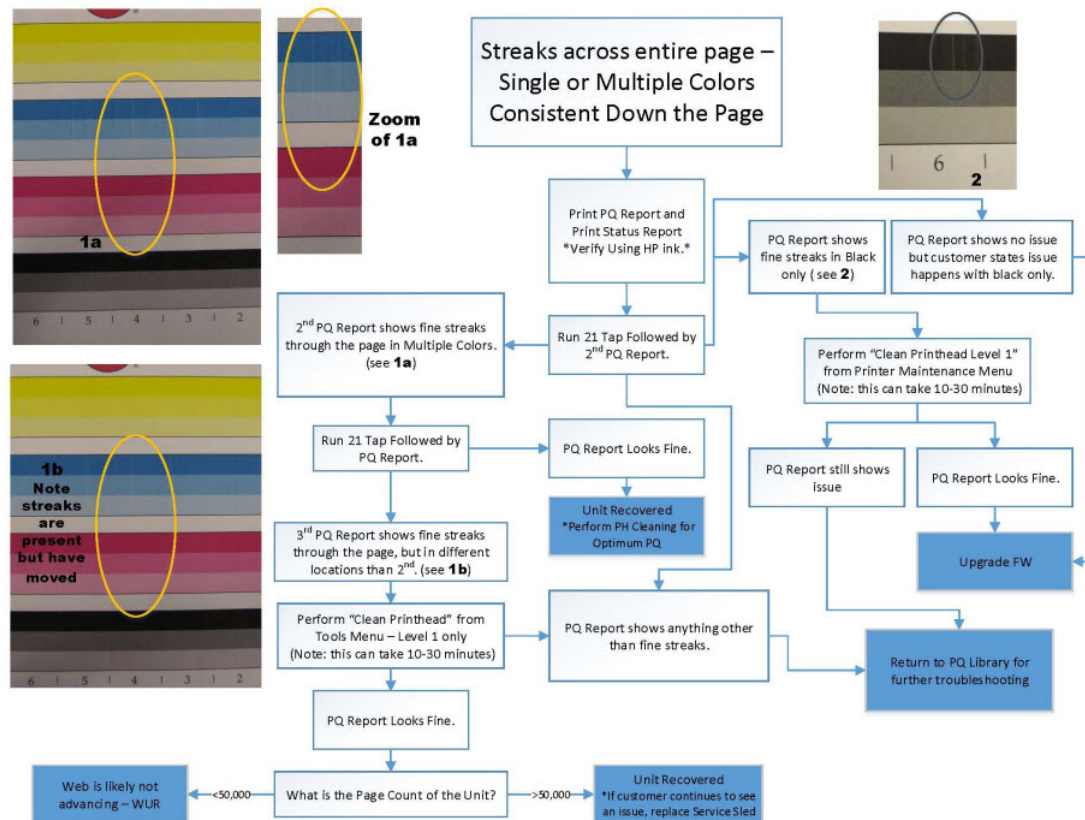
Click any of the following links to skip to the section in this document that contains steps to perform the specific task:

- [Print a Print Quality Report](#)
- [Print a Printer Status Report](#)
- [Run a 21 Tap Test](#)

If the print bar has failed, see the [Printbar Replacement Policy](#) (c04926672) for next steps.

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Streaks across entire page - Single or multiple colors consistent down the page



Possible causes:

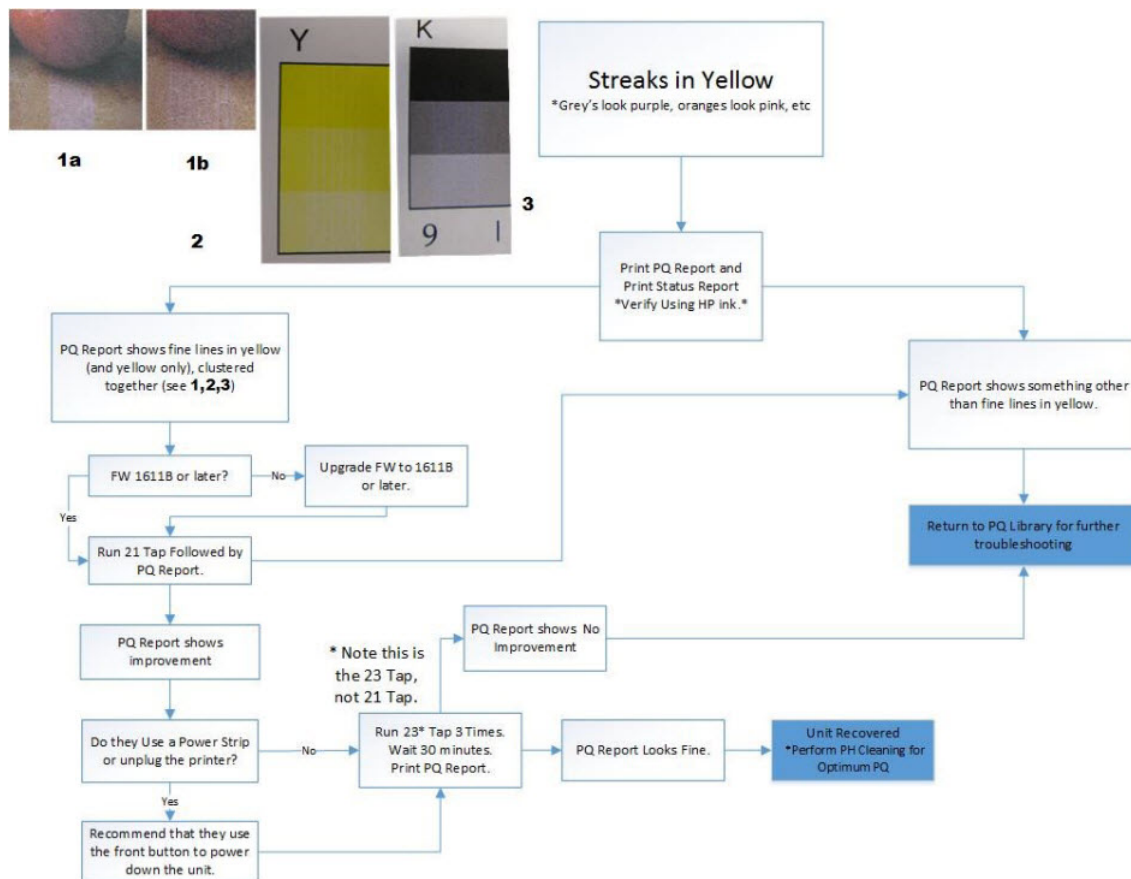
- **Non-recoverable:** Might have been caused by the web not advancing or a service sled failure.
- **Recoverable:**
 - In **older units**, a wipe can introduce air into the nozzles. Wipes are intended to occur when the unit is not in use, but sometimes happen when the customer needs to print. In these cases, the defect will clear if the unit is allowed to sit for 15 minutes.
 - **Black only defects** are different from ones affecting colors, and tend to appear when printing from slow devices, such as phones or networks. This defect has been mitigated with newer firmware.

Click any of the following links to skip to the section in this document that contains steps to perform the specific task:

- [Print a Print Quality Report](#)
- [Print a Printer Status Report](#)
- [Run a 21 Tap Test](#)

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Streaks in yellow

**Possible causes:**

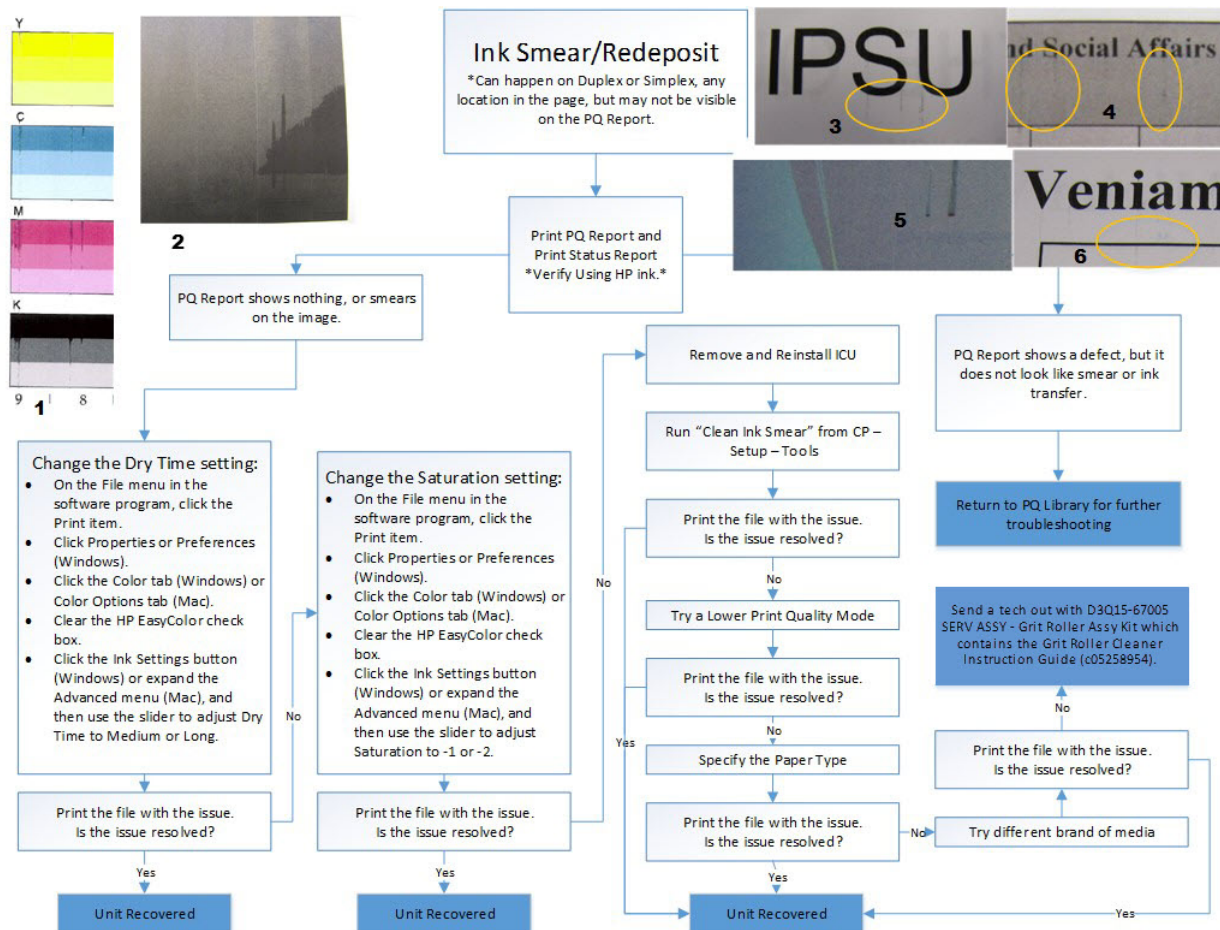
- **Recoverable:** Caused by ink buildup on the printbar (in older FW versions).
Update older firmware. An error in FW version 1548D prevented servicing for this defect if the customer turned the printer off. The result is ink buildup on the printbar. All printers with FW before 1611B should be updated and serviced to remove ink buildup. The defect should not reoccur once the firmware has been updated and the printer has been serviced.

Click any of the following links to skip to the section in this document that contains steps to perform the specific task:

- [Print a Print Quality Report](#)
- [Print a Printer Status Report](#)
- [Run a 21 Tap Test](#)
- [Run a 23 Tap Test](#)

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Ink smear



The solution order may be adapted to the priorities of the customer.

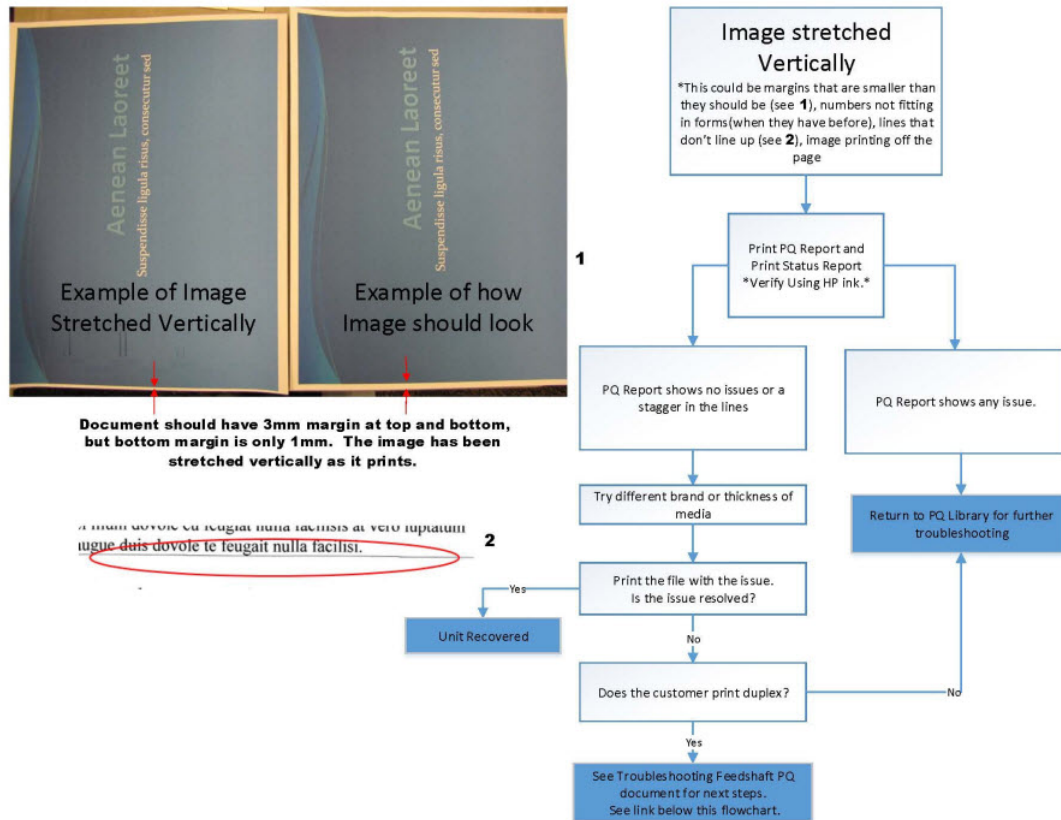
Increasing dry time sacrifices speed.

Click any of the following links to skip to the section in this document that contains steps to perform the specific task:

- [Print a Print Quality Report](#)
- [Print a Printer Status Report](#)

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Image stretched vertically



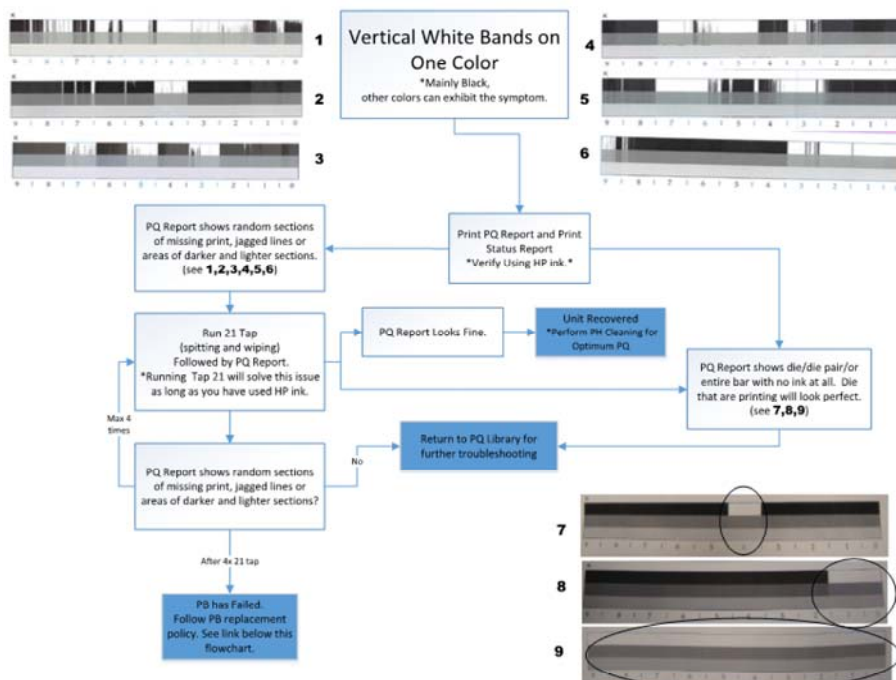
Click any of the following links to skip to the section in this document that contains steps to perform the specific task:

- [Print a Print Quality Report](#)
- [Print a Printer Status Report](#)

See [Troubleshooting Feedsheet Print Quality](#) (c04947362) for the next steps in troubleshooting this issue.

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Vertical white bands on one color



Click any of the following links to skip to the section in this document that contains steps to perform the specific task:

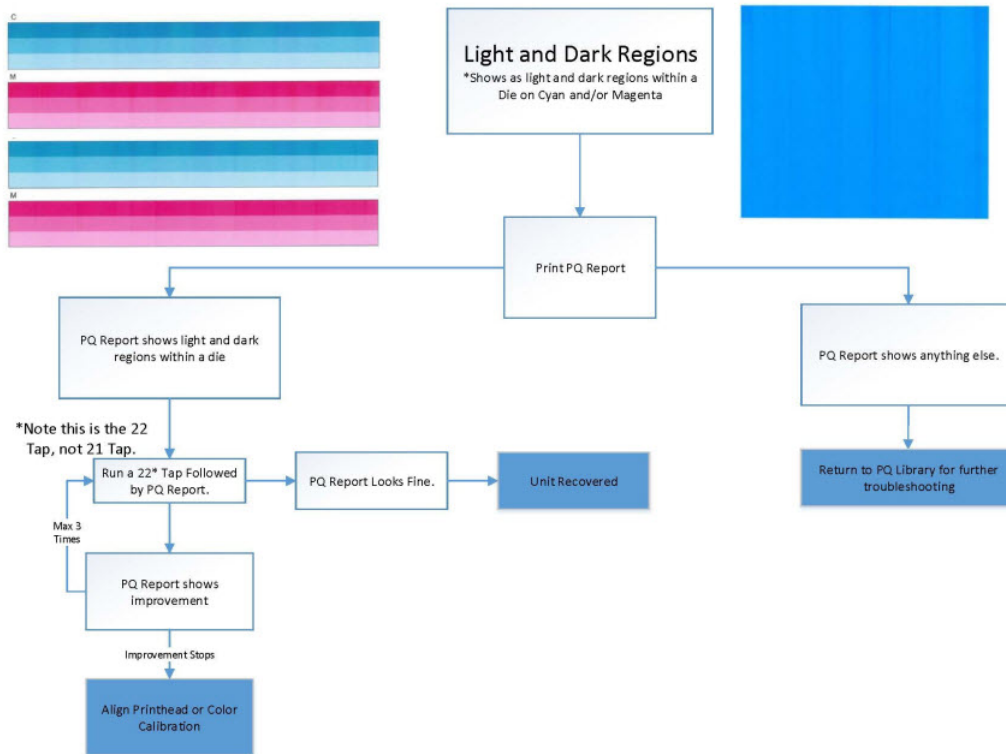
- [Print a Print Quality Report](#)
- [Print a Printer Status Report](#)

- [Run a 21 Tap Test](#)

If the print bar has failed, see the [Printbar Replacement Policy](#) (c04926672) for next steps.

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Light and dark regions



Click any of the following links to skip to the section in this document that contains steps to perform the specific task:

- [Print a Print Quality Report](#)
- [Print a Printer Status Report](#)
- [Run a 22 Tap Test](#)

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Columns of misaligned print

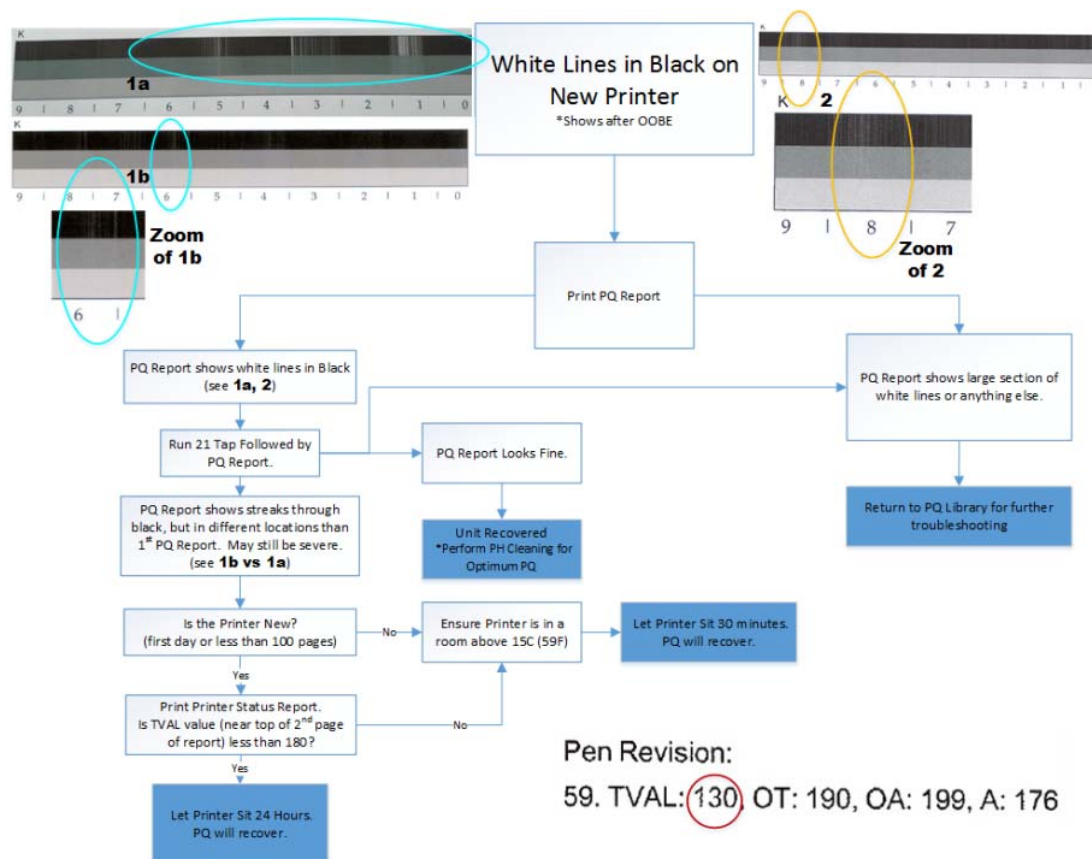
Please refer to the following document for steps on resolving this issue:

[Printed pages are misaligned in specific areas and show a pattern of rectangles](#) (c05270662)

If the steps in the document referenced above do not resolve the issue, replace the unit.

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White lines in black (on new printer)



Click any of the following links to skip to the section in this document that contains steps to perform the specific task:

- [Print a Print Quality Report](#)
- [Print a Printer Status Report](#)
- [Run a 21 Tap Test](#)

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Identify genuine HP ink and non-HP ink

To determine if the customer is using genuine HP ink, or ink from another manufacturer, follow the steps in the table below.

Example of a Printer Status Report

1: Example of Cartridge System Information on Printer Status Report (page 1)

43. USE:	0	0	0	0
46. CMA:	0	0	0	0
48. HP:	1	1	1	1
46. Approximate Pages Remaining:	4000	4000	4000	4000
47. Supported Cartridge(s):	HP 875A	HP 875A	HP 875A	HP 875A
	HP 875X	HP 875X	HP 875X	HP 875X
	HP 875Y	HP 875Y	HP 875Y	HP 875Y
48. NANO:	0	0	0	0
49. PPD:	0	0	0	0

**Not all cartridges are available in all regions.

2: Example of the Print Head Information section on the Printer Status Report (page 2)

Print Head Information

58. Registers:	65	64	63	62	61	60	59	58	57	56
1.	2b	2a	29	2a	2b	2a	2b	2a	2b	2a
2.	43	42	41	40	39	38	37	36	35	34
3.	35	5	15	5	15	15	15	15	15	15
4.	b	0	a	a	a	a	a	a	a	af
5.	26	20	26	26	26	26	26	26	41	41
6.	46	44	44	44	44	44	44	44	50	50
7.	0	3	0	0	3	0	3	0	2	3

59. Cartridge Information:	65	64	63	62	61	60	59	58	57	56
1.	2b	2a	29	2a	2b	2a	2b	2a	2b	2a
2.	43	42	41	40	39	38	37	36	35	34
3.	35	5	15	5	15	15	15	15	15	15
4.	b	0	a	a	a	a	a	a	a	af
5.	26	20	26	26	26	26	26	26	41	41
6.	46	44	44	44	44	44	44	44	50	50
7.	0	3	0	0	3	0	3	0	2	3

Determine if genuine HP ink or another brand is being used

HP recommends that you use genuine HP ink supplies.

Learn how to identify genuine HP cartridges by the security label on the box.

3: Example of security label on genuine HP cartridges

Visit www.hp.com/go/anticounterfeit to learn more.

1. Print a Printer Status Report.
2. Look at the Printer Status Report and locate the following line-items:

These line-items will contain values that can be used to determine whether or not a non-HP brand of ink has been used in the printer. Refer to the sample Printer Status Report (pictured on the left).

The following line-items can be found in the **Cartridge System Information** section on the Printer Status Report (pictured on the left).

- **USE:**
If this value is 0, then genuine HP ink is being used.
If this value is 1, then a different brand of ink is being used.
- **HP:**
If this value is 1, then genuine

1 - Arrow points to column 1, line 4 The Printer Status Report might look slightly different than the example above (depending on your product model and firmware version).	HP ink is being used. If this value is 0, then a different brand of ink is being used. • NHIO: If this value is 0, then only genuine HP ink has been used. If this value is 1, then a different brand of ink was used in the past. • IRO: If this value is 0, then only genuine HP ink has been used. If this value is 1, then a refill has been used in the past. The following line-items can be found in the Print Head Information section on the Printer Status Report (pictured on the left). • Registers: Locate the value in column 1, line 4 (see the arrow pictured to the left). This value will contain two characters. If the first character is 8, 9, a, b, c, d, e, or f, then a different brand of ink has been used at some point in this printbar.
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Obtain a Printer Status Report, Print Quality Report, and run a Tap Test

The sections below contain steps on how to print a Printer Status Report and a Print Quality Report, and also the steps to run the various Tap Tests.

IMPORTANT:

All of the PQ troubleshooting steps in this document will require you to print these reports and run a tap test. **These reports and the tap test can all be accessed from the Reports Menu that is part of the hidden Support Menu on the product control panel.**

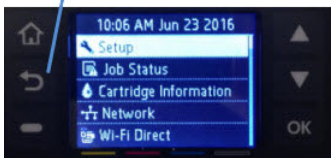
The steps for accessing the hidden Support Menu and the Reports Menu are included in the sections below.

Once you have reached the Reports Menu, you can use the arrows to toggle between the different options. This way you will not have to return to the Home screen when you are attempting to access any of these options. Keep this in mind when helping a customer obtain these reports and run the tap test.

The steps contained in this document for accessing the Printer Status Report and the Print Quality Report are different from the steps for accessing these reports through the Setup Menu. These steps are intended for use by call center agents to facilitate troubleshooting.

Obtain a Printer Status Report

Find your printer type (monochrome or color control panel display) in the table below and then follow the steps provided to print a Printer Status Report.

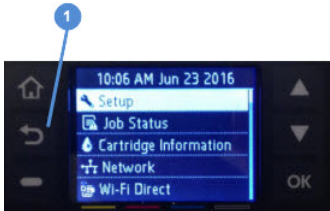
Monochrome control panel display	Color control panel display
- From the Home screen on the printer control panel, press the Back (↩) button four (4) times. 4: Example of the Home screen  1 - Back button The Support menu will display. - Press the arrow buttons (▼ or ▲) until Reports Menu is highlighted, and then press the OK button. - Press the arrow buttons (▼ or ▲) until Printer Status Report is highlighted, and then press the OK button. The Printer Status Report will print.	- From the Home screen on the printer control panel, touch the Back (↩) button four (4) times. The Back (↩) button will not be illuminated. Press the control panel where the button would normally appear illuminated. 5: Example of the Home screen where the Back button would normally be illuminated (circled area)  The Support menu will display. - Scroll to and touch Reports Menu. The Reports menu will display. - Scroll to and touch Printer Status Report, and then touch OK. The Printer Status Report will print.

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Obtain a Print Quality Report

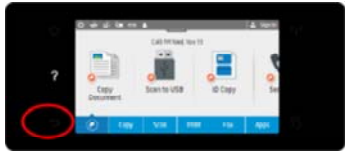
Find your printer type (monochrome or color control panel display) in the table below and then follow the steps provided to print a Print Quality Report.

Monochrome control panel display	Color control panel display
From the Home screen on the printer control panel, press the Back (↩) button four (4) times. 6: Example of the Home screen	From the Home screen on the printer control panel, touch the Back (↩) button four (4) times. The Back (↩) button will not be illuminated. Press the control panel where the button would normally appear illuminated. 7: Example of the Home screen where the Back button would normally be illuminated (circled area)



1 - Back button
The Support menu will display.

- Press the arrow buttons (▼ or ▲) until Reports Menu is highlighted, and then press the OK button.
- Press the arrow buttons (▼ or ▲) until Print Quality Report is highlighted, and then press the OK button.
The Print Quality Report will print.



The Support menu will display.

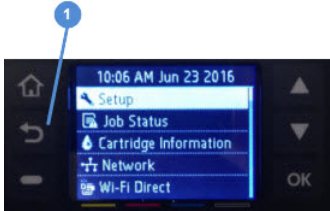
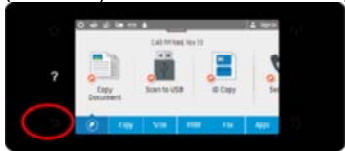
- Scroll to and touch Reports Menu.
The Reports menu will display.
- Scroll to and touch Print Quality Report, and then touch OK.
The Print Quality Report will print.

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Run a 21 Tap Test

Find your printer type (monochrome or color control panel display) in the table below and then follow the steps provided to run a 21 Tap Test.

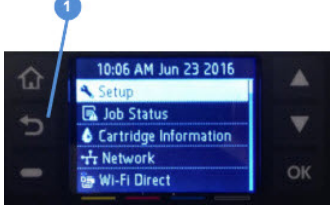
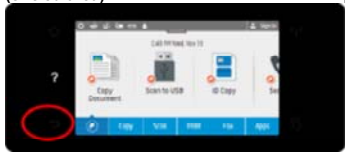
This tap test takes approximately 8 minutes.

Monochrome control panel display	Color control panel display
- From the Home screen on the printer control panel, press the Back (↵) button four (4) times. 8: Example of the Home screen  1 - Back button The Support menu will display. - Press the arrow buttons (▼ or ▲) until Reports Menu is highlighted, and then press the OK button. - Press the arrow buttons (▼ or ▲) until Print-mech button tap is highlighted, and then press the OK button. - Press the Up arrow (▲) until 21 displays, and then press the OK button. The printer will run the 21 Tap Test.	- From the Home screen on the printer control panel, touch the Back (↵) button four (4) times. The Back (↵) button will not be illuminated. Press the control panel where the button would normally appear illuminated. 9: Example of the Home screen where the Back button would normally be illuminated (circled area)  The Support menu will display. - Scroll to and touch Reports Menu. The Reports menu will display. - Scroll to and touch Print-mech button tap, and then touch OK. - Touch the Up arrow (▲) until 21 displays, and then touch OK. The printer will run the 21 Tap Test.

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Run a 22 Tap Test

Find your printer type (monochrome or color control panel display) in the table below and then follow the steps provided to run a 22 Tap Test.

Monochrome control panel display	Color control panel display
- From the Home screen on the printer control panel, press the Back (↵) button four (4) times. 10: Example of the Home screen  1 - Back button The Support menu will display. - Press the arrow buttons (▼ or ▲) until Reports Menu is highlighted, and then press the OK button. - Press the arrow buttons (▼ or ▲) until Print-mech button tap is highlighted, and then press the OK button. - Press the Up arrow (▲) until 22 displays, and then press the OK button. The printer will run the 22 Tap Test.	- From the Home screen on the printer control panel, touch the Back (↵) button four (4) times. The Back (↵) button will not be illuminated. Press the control panel where the button would normally appear illuminated. 11: Example of the Home screen where the Back button would normally be illuminated (circled area)  The Support menu will display. - Scroll to and touch Reports Menu. The Reports menu will display. - Scroll to and touch Print-mech button tap, and then touch OK. - Touch the Up arrow (▲) until 22 displays, and then touch OK. The printer will run the 22 Tap Test.

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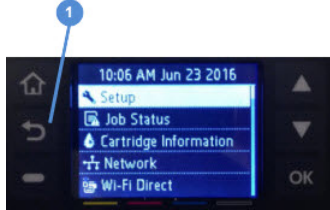
Run a 23 Tap Test

Find your printer type (monochrome or color control panel display) in the table below and then follow the steps provided to run a 23 Tap Test.

Monochrome control panel display	Color control panel display
From the Home screen on the printer control	From the Home screen on the printer control

panel, press the Back (↩) button four (4) times.

12: Example of the Home screen



1 - Back button

The Support menu will display.

2. Press the arrow buttons (▼ or ▲) until Reports Menu is highlighted, and then press the OK button.
3. Press the arrow buttons (▼ or ▲) until Print-mech button tap is highlighted, and then press the OK button.
4. Press the Up arrow (▲) until 23 displays, and then press the OK button.
The printer will run the 23 Tap Test.

panel, touch the Back (↩) button four (4) times.

The Back (↩) button will not be illuminated. Press the control panel where the button would normally appear illuminated.

13: Example of the Home screen where the Back button would normally be illuminated (circled area)



The Support menu will display.

2. Scroll to and touch Reports Menu.
The Reports menu will display.
3. Scroll to and touch Print-mech button tap, and then touch OK.
4. Touch the Up arrow (▲) until 23 displays, and then touch OK.
The printer will run the 23 Tap Test.

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